

CYBERBULLYING DETECTION ON SOCIAL MEDIA USING MACHINE LEARNING

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ABSTRACT

This study compares SVM, Random Forest, and LSTM classifiers for detecting cyberbullying in Filipino-English social media content, finding that SVM with TF-IDF features achieves the best F1-score of 88.3%.

INTRODUCTION

Cyberbullying is a growing issue among Filipino youth, exacerbated by high social media usage rates. Platforms lack automated Filipino-language moderation. NLP-based detection can flag harmful content in real time. This study builds and compares classifiers on a dataset of 12,000 annotated Filipino-English (Taglish) social media posts.

METHODOLOGY

A dataset of 12,000 Taglish tweets and Facebook comments was collected and annotated into Bullying and Non-Bullying classes (60/40 split). Three classifiers were compared: SVM with TF-IDF (unigrams+bigrams), Random Forest with word2vec embeddings, and a Bidirectional LSTM. Preprocessing: lowercasing, stopword removal, code-mixed text normalization, and emoji-to-text conversion. Evaluation: F1-score, precision, recall, and 5-fold cross-validation.

RESULTS

SVM achieved the highest F1-score of 88.3% (precision 89.1%, recall 87.6%). Random Forest scored F1=85.7% and BiLSTM scored F1=87.1%. SVM training time was 12 seconds vs. LSTM's 4 hours. SVM misclassified sarcastic expressions as non-bullying in 7.4% of cases.

DISCUSSION

SVM with TF-IDF remains a strong baseline for short-text classification, outperforming even deep learning on this dataset size. Sarcasm and code-mixing are the primary error sources. The class imbalance (60/40) was addressed with SMOTE oversampling. Future work should explore mBERT and XLM-R for multilingual Filipino-English understanding.